Amujuri Sai Praveen

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Applied AI & Full Stack Engineer building production-ready AI-powered systems. Experienced in integrating machine learning models into scalable web applications using modern backend and cloud deployment tools.

Education

ANITS College-CGPA:7.62 — Visakhapatnam, AP
Bachelor of Technology in Artificial Intelligence & Machine Learning — *2024 – 2027 (Expected)*

Andhra Polytechnic College-CGPA:7.89 — Kakinada, AP
Diploma in Computer Science & Engineering — *2020 –2024*

Experience

SkillDzire — **May 2025 - Jun 2025**
Machine Learning Intern — *Hyderabad (Remote)*

- Implemented **end-to-end ML pipelines** using **CNNs** for image classification and **TF-IDF-based NLP** for content recommendation systems.
- Optimized structured dataset models (**Random Forest**, **SVM**) through **hyperparameter tuning**, achieving **~92% accuracy**.
- Engineered **data preprocessing pipelines** using **Pandas** and **NumPy** for feature normalization and model readiness.

Krify — **Aug 2022 - Jan 2023**
Full Stack Developer Intern — *Kakinada (Onsite)*

- Built a full-stack web application under senior mentorship using the **MERN stack (React, Node.js, Express.js, MongoDB).**
- Developed **RESTful APIs** to enable communication between frontend components and the MongoDB database.
- Designed **MongoDB schemas** and implemented **CRUD operations** to manage application data efficiently.
- Implemented **JWT-based authentication** and basic authorization mechanisms to secure user access.
- Participated in code reviews and collaborative debugging sessions within an **Agile-style development workflow**.

Projects

LunaBot – Autonomous Lunar Rover Prototype [GitHub]
Unity | ROS Noetic | Python | C# | YOLOv8 | RTAB-Map | Flask | WebSockets

- Built an autonomous robotics pipeline integrating **Unity (simulation), ROS Noetic, YOLOv8, and RTAB-Map** for end-to-end navigation and semantic mapping.
- Engineered a custom C# transform bridge converting Unity's left-handed system to **REP-103 compliant ROS frames**, ensuring stable SLAM and accurate TF hierarchy.
- Simulated real-time sensors (180-ray LiDAR, 32FC1 depth, RGB, odometry) and streamed ROS topics via **rosbridge WebSockets**.
- Developed a low-latency **Flask + Socket.IO dashboard**, reducing SLAM map payloads from ~2MB to <50KB and maintaining <50ms end-to-end latency.
- Achieved **12–15ms YOLOv8 inference time per frame** while maintaining full autonomous operation at only **~10–15% CPU load**.

ZeroTouch – Touchless Medical Interface Prototype [GitHub]
Python | OpenCV | MediaPipe | NumPy | Tkinter | Computer Vision | Human-Computer Interaction (HCI)

- Architected a contactless **DICOM interface** for sterile environments, mitigating contamination risks via discrete hand gestures.
- Engineered a vector-based gesture engine to distinguish intentional swipes from noise using **landmark velocity (<40ms latency).**
- Integrated **CLAHE preprocessing** to normalize extreme lighting (10–90% luminance) for robust detection on gloves and bare skin.
- Implemented **temporal consistency checks**, reducing false positives by **~90%** to prevent accidental inputs during procedures.
- Mapped **Euclidean distance heuristics** to responsive UI controls (Swipe, Zoom) for jitter-free interaction.

MediStream AI – AI-Powered Hospital Booking Platform [GitHub] [Live Demo]
React | FastAPI | Express | MongoDB | Gemini API | JWT | Vercel | Render

- Developed a full-stack hospital triage platform integrating **Google Gemini (LLM)** for automated symptom analysis and urgency scoring (1–10 scale).
- Implemented secure **JWT-based authentication** and designed role-based access control for patients and administrators.
- Designed a dynamic priority queue system to reorder appointments in real time, ensuring high-urgency cases are scheduled first.
- Deployed a decoupled **REST API** on Render and frontend on Vercel, enabling scalable client-server communication.

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript
Frontend: React, Next.js, HTML, CSS, Tailwind CSS, Material UI
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, JWT Authentication, Socket.io
Databases & Messaging: MongoDB, Firebase, SQL
Deployment & Tools: AWS EC2, Git, GitHub, Docker (basic), Vercel, Render

Certifications

Fundamentals of Accelerated Computing with CUDA Python – **NVIDIA DLI**
Postman Student Expert **– Postman**